

$$1.1 \quad \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$\begin{array}{l} \text{2 def. normy} \quad \text{2 axiomu skalárního součinu o linearitě (2.)} \\ \downarrow \quad \downarrow \\ \|\vec{u} + \vec{v}\|^2 = \langle \vec{u} + \vec{v}, \vec{u} + \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle \end{array}$$

$$\begin{aligned} x + \bar{x} &= \operatorname{Re}(x) + i\operatorname{Im}(x) + \operatorname{Re}(x) - i\operatorname{Im}(x) = \\ &= 2\operatorname{Re}(x) \end{aligned}$$

$$\|\vec{u} - \vec{v}\|^2 = \langle \vec{u} - \vec{v}, \vec{u} - \vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, \vec{v} \rangle - \langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle$$

$$\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2 = 2(\langle \vec{u}, \vec{v} \rangle + \langle \vec{v}, \vec{u} \rangle) = 2(\langle \vec{u}, \vec{v} \rangle + \overline{\langle \vec{u}, \vec{v} \rangle}) = 2(2\operatorname{Re}(\langle \vec{u}, \vec{v} \rangle))$$

$$\Rightarrow \operatorname{Re}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} + \vec{v}\|^2 - \|\vec{u} - \vec{v}\|^2)$$

$$\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$

$$\begin{array}{l} \text{2 def. normy} \quad \text{2 axiomu skalárního součinu o linearitě (2.)} \\ \downarrow \quad \downarrow \\ \|\vec{u} - i\vec{v}\|^2 = \langle \vec{u} - i\vec{v}, \vec{u} - i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle - \langle \vec{u}, i\vec{v} \rangle - \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \\ \text{2 axiomu skalárního součinu o linearitě (1.)} \quad \text{a z pozorování } \langle i\vec{u}, \vec{v} \rangle = i\langle \vec{u}, \vec{v} \rangle \\ = \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle + i^2 \langle \vec{v}, \vec{v} \rangle = \\ = \langle \vec{u}, \vec{u} \rangle - i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ = \|\vec{u}\|^2 + \|\vec{v}\|^2 - i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) = \\ = \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{array}$$

$$\begin{aligned} x - \bar{x} &= \operatorname{Re}(x) + i\operatorname{Im}(x) - (\operatorname{Re}(x) - i\operatorname{Im}(x)) = \\ &= 2i\operatorname{Im}(x) \end{aligned}$$

$$\begin{array}{l} \|\vec{u} + i\vec{v}\|^2 = \langle \vec{u} + i\vec{v}, \vec{u} + i\vec{v} \rangle = \langle \vec{u}, \vec{u} \rangle + \langle \vec{u}, i\vec{v} \rangle + \langle i\vec{v}, \vec{u} \rangle + \langle i\vec{v}, i\vec{v} \rangle = \\ \text{2 axiomu skalárního součinu o linearitě (1.)} \quad \text{a z pozorování } \langle i\vec{u}, \vec{v} \rangle = i\langle \vec{u}, \vec{v} \rangle \\ = \langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle + i\langle \vec{v}, \vec{u} \rangle + i^2 \langle \vec{v}, \vec{v} \rangle = \\ = \langle \vec{u}, \vec{u} \rangle + i\langle \vec{u}, \vec{v} \rangle - i\langle \vec{v}, \vec{u} \rangle + \langle \vec{v}, \vec{v} \rangle = \\ = \|\vec{u}\|^2 + \|\vec{v}\|^2 + i(\langle \vec{u}, \vec{v} \rangle - \overline{\langle \vec{u}, \vec{v} \rangle}) = \\ = \|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{array}$$

$$\begin{aligned} x - \bar{x} &= \operatorname{Re}(x) + i\operatorname{Im}(x) - (\operatorname{Re}(x) - i\operatorname{Im}(x)) = \\ &= 2i\operatorname{Im}(x) \end{aligned}$$

$$\begin{aligned} \|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2 &= \|\vec{u}\|^2 + \|\vec{v}\|^2 + 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) - (\|\vec{u}\|^2 + \|\vec{v}\|^2 - 2\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle)) = \\ &= 4\operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) \end{aligned}$$

$$\Rightarrow \operatorname{Im}(\langle \vec{u}, \vec{v} \rangle) = \frac{1}{4} (\|\vec{u} - i\vec{v}\|^2 - \|\vec{u} + i\vec{v}\|^2)$$